

Short peptides self-assemble to produce catalytic amyloids

Caroline M. Rufo^{1†}, Yurii S. Moroz^{1†}, Olesia V. Moroz^{1†}, Jan Stöhr², Tyler A. Smith¹, Xiaozhen Hu³, William F. DeGrado^{3*} and Ivan V. Korendovych^{1*}

Enzymes fold into unique three-dimensional structures, which underlie their remarkable catalytic properties. The requirement to adopt a stable, folded conformation is likely to contribute to their relatively large size (>10,000 Da). However, much shorter peptides can achieve well-defined conformations through the formation of amyloid fibrils. To test whether short amyloid-forming peptides might in fact be capable of enzyme-like catalysis, we designed a series of seven-residue peptides that act as Zn²⁺-dependent esterases. Zn²⁺ helps stabilize the fibril formation, while also acting as a cofactor to catalyse acyl ester hydrolysis. These results indicate that prion-like fibrils are able to not only catalyse their own formation, but they can also catalyse chemical reactions. Thus, they might have served as intermediates in the evolution of modern-day enzymes. These results also have implications for the design of self-assembling nanostructured catalysts including ones containing a variety of biological and non-biological metal ions.

The mechanisms by which proteins evolved to possess enzymatic activity are largely unresolved. An enzyme's amino-acid sequence underlies its ability to fold into a native structure, which positions and tunes the dynamics of the functional groups required for binding and catalysis. Modern-day enzymes are generally at least 100 residues in length, which is close to the chain length required to create a well-defined hydrophobic core in a globular protein. A protein of this length, however, would have an astronomically large number of possible sequences, and it is unclear how nature might have tested sufficient numbers of permutations to find one capable of folding and function.

This apparent paradox has led to the suggestion that the earliest proteins might have been formed from short peptides or polymers with repeating amino-acid sequences that relied on self-assembly to achieve a folded structure^{1,2}. Indeed, simple heptapeptides with alternating apolar and polar residues self-assemble into extended β sheets stabilized through intermolecular association of the apolar residues in the sequence repeat³. Moreover, structural characterization of short amyloid-forming peptides has revealed a myriad of conformational variations on the basic cross- β conformation^{4–6}. Thus, it is possible that the first primitive protein enzymes were short peptides with amyloid structures providing the frameworks to support their catalytic activities.

Metals also might have figured largely in the emergence of the catalytic activity of proteins^{7,8}. Not only do transition-metal ions help catalyse reactions, but they can also influence the folding of proteins through interaction with ligating side chains. We therefore explored the design, metal interactions and catalytic properties of Zn²⁺-binding amyloidogenic peptides. Zn²⁺ was chosen as a cofactor given its high abundance and frequent occurrence in modern metal-dependent hydrolases such as carbonic anhydrase and metalloproteases. The Zn²⁺ ion lowers the pK_a of a bound water, stabilizing and positioning hydroxide for nucleophilic attack of the substrate. The Zn²⁺ binding site in enzymes often contains two His ligands projecting from positions *i* and *i* + 2 of a β -sheet; a third His ligand from a neighbouring strand completes the

primary ligand environment. This very simple interstrand 3-His metal-binding site is observed in proteins such as erythrocyte carbonic anhydrase (Fig. 1a)⁹; similarly, two His residues from neighbouring positions of β -strands are seen in the amyloid-like β -solenoid structure of an archaeal γ -carbonic anhydrase¹⁰. Additionally, general acid/bases from neighbouring side chains serve as secondary ligands and assist protonation/deprotonation events in catalysis. We therefore asked whether a similar site might be created within the β -sheet-forming minimalist heptapeptide sequence (LKLKLLK). This and related peptides have alternating hydrophobic residues, which creates an amphiphilic β -strand that further associates via hydrophobic interactions to form extended amyloid β -structures^{11,12}. Considerable sequence variation is tolerated as long as the hydrophobic periodicity is preserved^{13,14}. We retained the apolar Leu residues to drive assembly, while the Lys side chains at positions 2, 4 and 6 were changed to polar residues capable of supporting transition-metal ion binding and catalysis. Zn²⁺-binding His residues were placed at positions 2 and 4, while at position 6 (Fig. 1b) we explored acidic (Asp, Glu), neutral (Gln, Tyr) and basic residues with various pK_a values (His, Lys, Arg).

Results and discussion

Hydrolytic enzymes perform a diverse set of functions ranging from amide, ester or lipid hydrolysis to catalysing the equilibration between CO₂ and bicarbonate. Despite such diversity, many hydrolytic enzymes are assayed and benchmarked using a simple chromogenic substrate, *p*-nitrophenylacetate (*p*NPA). We therefore chose this substrate to allow for comparison with natural enzymes and various designed protein catalysts. Most of the seven original designs displayed measurable activity over the buffer control (pH 8 in the presence of 1 mM ZnCl₂) in the initial screening for *p*NPA hydrolysis, as shown in Table 1. The dependence of the initial rate of the reaction on substrate concentration for a typical peptide (7, LHLHLRL) is consistent with the Michaelis–Menten model, with $k_{\text{cat}} = 3.2 \pm 0.4 \times 10^{-2} \text{ s}^{-1}$, $K_{\text{M}} = 1.8 \pm 0.4 \text{ mM}$ and a catalytic efficiency of $k_{\text{cat}}/K_{\text{M}} = 18 \pm 4 \text{ M}^{-1} \text{ s}^{-1}$ (Fig. 2a, Table 1).

¹Department of Chemistry, Syracuse University, Syracuse, New York 13244, USA, ²Institute for Neurodegenerative Diseases and Department of Neurology, University of California – San Francisco, San Francisco, California 94143, USA, ³Department of Pharmaceutical Chemistry, University of California – San Francisco, San Francisco, California 94158, USA; [†]These authors contributed equally to this work. *e-mail: ikorendo@syr.edu; william.degrado@ucsf.edu

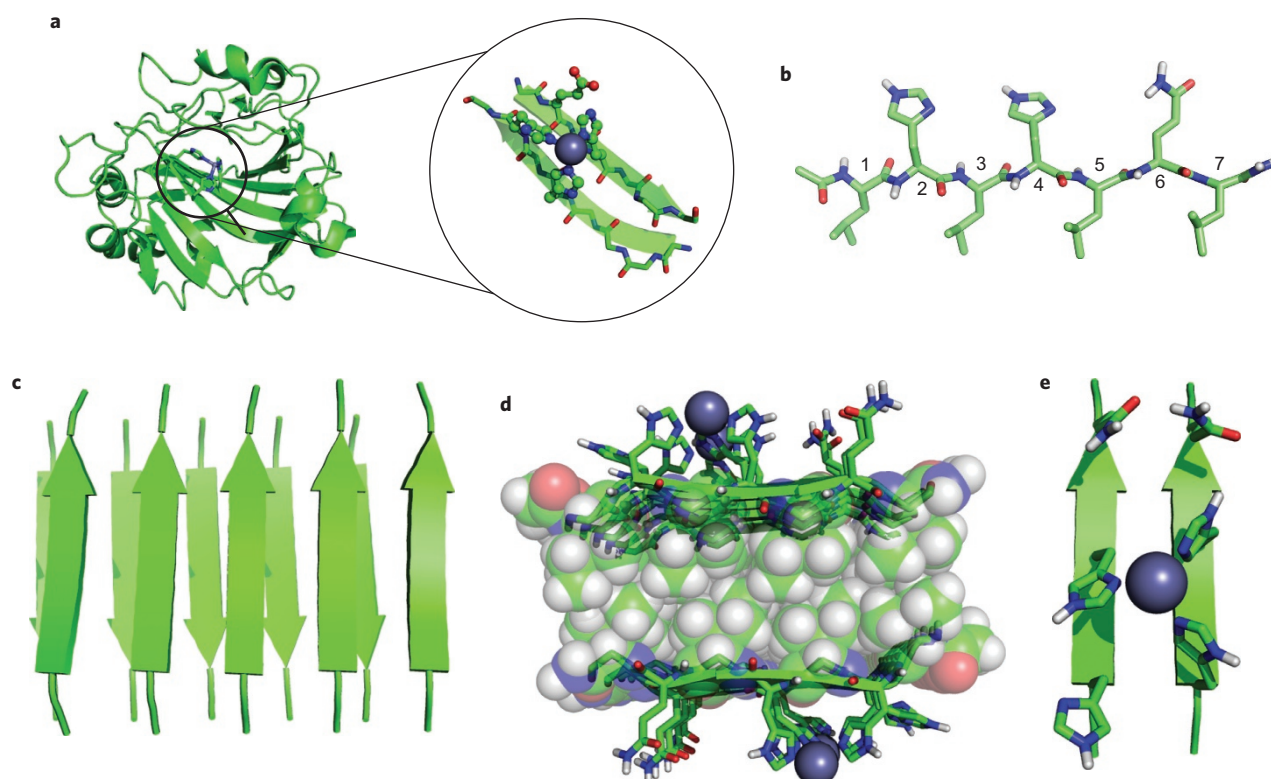


Figure 1 | Overview of concept and design. **a**, Structure of human carbonic anhydrase showing a typical metal-binding motif. **b**, Model for one of the designed peptides (**11**, Ac-IHIHIQI-CONH₂) in the extended β -strand configuration showing positions of the residues in the sequence. **c–e**, Computationally derived model of fibrils formed by **11**, showing overall fold (**c**), packing of the hydrophobic core (**d**) and zinc primary coordination sphere (**e**).

The activity of **7** is strictly zinc-dependent, and no significant activity above background is observed in the absence of Zn²⁺. The catalytic activity also showed a marked dependence on the nature of the residue at position 6. Interestingly, peptide **3**, with an amyloidogenic glutamine residue at this position was most active, with $k_{\text{cat}}/K_M = 30 \pm 3 \text{ M}^{-1} \text{ s}^{-1}$. Peptides with Asp, Glu or His residues at position 6 (**1**, **2** and **5**, respectively) had little to no activity,

possibly because these side chains compete with the His residues at positions 2 and 4 for binding to the metal ion creating catalytically inactive sites.

Structure–activity relationships. Intrigued by the finding that a potentially amyloid-promoting Gln residue enhanced catalytic activity we varied the hydrophobic Leu residues that were expected to stabilize the β -sheet association. Replacement of the Leu residues with residues of greater β -sheet propensity^{15,16}, Ile or Val, produced peptides **9** (IHIHIRI) and **10** (VHVHVRV) with almost 50% increase in activity compared to **7**. Conversely, replacement of Leu with Ala, a residue with both lower hydrophobicity and weaker β -sheet-forming propensity, gave a peptide (**8**) that is almost inactive as an esterase. The terminal acetyl and carboxamide groups also appeared to be important for activity, because the peptide lacking these groups (**14**) was catalytically inactive. These observations are in good agreement with previous studies that showed that subtle modifications of secondary structure in short peptide catalysts lead to substantial improvements in activity and substrate specificity^{17,18}.

The activity of **9** (IHIHIRI) could be further increased by inclusion of Gln at position 6, which had already been shown to increase activity in the leucine-containing peptides. The effects of introducing β -branched amino acids and glutamine are additive: the peptide containing both isoleucines in its hydrophobic core and Gln at position 6 (**11**, IHIHIQI) is 3.5-fold more active than **7** (Table 1) and reaches a k_{cat}/K_M value of $360 \pm 30 \text{ M}^{-1} \text{ s}^{-1}$ at pH 10.3 (Fig. 2b). The catalysis follows Michaelis–Menten kinetics, and no product inhibition was observed over 20 turnovers. A similar effect, albeit to a lower extent, was observed when Gln was introduced into the valine-containing peptide **10**. Interestingly, in both cases the gain in catalytic efficiency comes from the decrease in K_M rather than improvement in k_{cat} . The activity of **11** is more

Table 1 | Esterase activity for the designed peptides at pH 8 in the presence of 1 mM Zn²⁺.

Peptide	Sequence	k_{cat}/K_M ($\text{M}^{-1} \text{ s}^{-1}$)	$k_{\text{cat}} \times 10^{-2}$ (s^{-1})	K_M (mM)
Position 6 variants				
1	Ac-LHLHLDL-CONH ₂	0.2 ± 0.1		
2	Ac-LHLHLEL-CONH ₂	<0.2		
3	Ac-LHLHLQL-CONH ₂	30 ± 3		
4	Ac-LHLHLYL-CONH ₂	13 ± 5		
5	Ac-LHLHLHL-CONH ₂	0.6 ± 0.08		
6	Ac-LHLHLKL-CONH ₂	12 ± 2		
7	Ac-LHLHLRL-CONH ₂	18 ± 4		
Leucine substitutions				
7	Ac-LHLHLRL-CONH ₂	18 ± 4	3.2 ± 0.4	1.8 ± 0.4
8	Ac-AHAHARA-CONH ₂	0.12 ± 0.8		
9	Ac-IHIHIRI-CONH ₂	22 ± 8	4.2 ± 0.8	1.9 ± 0.5
10	Ac-VHVHVRV-CONH ₂	26 ± 4	3.8 ± 0.3	1.5 ± 0.2
Combined substitutions				
11	Ac-IHIHIQI-CONH ₂	62 ± 2	2.6 ± 0.04	0.4 ± 0.1
11a	Ac-VHVHVQV-CONH ₂	32 ± 2	1.6 ± 0.2	0.5 ± 0.1
Primary ligand variants				
12	Ac-IAIHIRI-CONH ₂	0.36 ± 0.16		
13	Ac-IHIAIRI-CONH ₂	0.2 ± 0.4		
Removal of N, C-terminal blocking groups				
14	H ₂ N-IHIHIQI-COOH	1 ± 3		

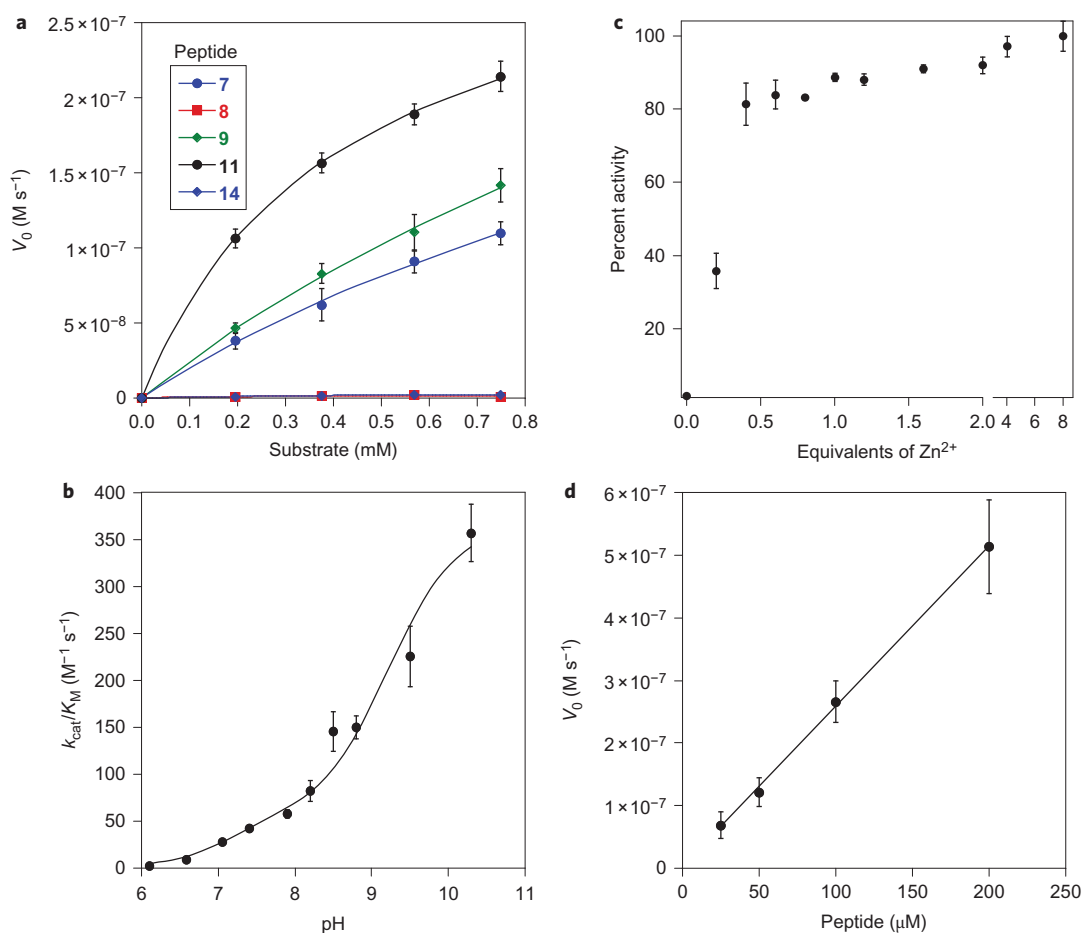


Figure 2 | Functional characterization of the catalysts. **a**, Plots for esterase activity catalysed by representative peptides (7, 8, 9, 11 and 14 as specified in the key) at pH 8 in the presence of 1 mM Zn^{2+} , with the smooth curve showing the fit to the Michaelis-Menten equation. The profile of the most active peptide (11) shows an enzyme-like saturation at higher substrate concentrations. **b**, The pH profile of the esterase activity of 11 is consistent with a two-state protonation model. **c**, Dependence of activity of 11 on the concentration of Zn^{2+} , showing a 2:1 peptide:metal stoichiometry. **d**, Dependence of *p*-nitrophenyl acetate (pNPA) hydrolysis rate catalysed by 11 (Ac-IHIIHQI-CONH₂) on the concentration of monomeric peptide used to make fibrils. The catalytic efficiency of the fibrils does not depend on the initial concentration of the monomeric peptide used for assembly. The reaction is first order in peptide in the concentration range studied. Error bars in all graphs represent standard deviations of at least three independent measurements.

than 100-fold greater in the presence than the absence of Zn^{2+} , showing that the activity does not arise from simple imidazole catalysis (which is considerably lower^{19,20} than the rate measured for Zn^{2+} -11 fibrils). Also, the activity of fibrils formed by 11 is more than tenfold greater than those of non-metal-based esterases identified in combinatorial libraries by Hecht *et al.*^{21,22} and designed by Baltzer *et al.*²³ and Mayo and colleagues²⁰. Fibrils formed by 11 are over an order of magnitude more active than *de novo* designed, zinc-binding trimeric coiled coils reported by Pecoraro *et al.*²⁴ and are on par with the most active (to date) zinc-protein complex reported by Kuhlman and co-workers²⁵. Moreover, the activity of 11 on a weight basis is greater than that of carbonic anhydrase, which catalyses pNPA hydrolysis with $k_{cat}/K_M = 2,550 M^{-1} s^{-1}$ (ref. 26). Such remarkable efficiency is due to the small size of the active unit in 11 (probably a dimer of seven-residue peptides), while the protein is at least 15-fold larger in molecular weight.

The proper coordination of the metal ion is also crucial, because a single mutation of either histidine residue to alanine results in at least 60-fold decrease in esterase activity (Table 1). Further examination of the catalytic activity of 11 (IHIIHQI) as a function of Zn^{2+} concentration showed that the activity increased linearly until a stoichiometry of ~1:2 (zinc to peptide) was reached, after which no significant increase was observed (Fig. 2c). This finding is consistent with the design in which two peptides cooperate to form the

active catalyst. The pH activity profile of 11 is consistent with two protonation/deprotonation steps with pK_a values of 7.1 ± 0.4 and 9.3 ± 0.1 (Fig. 2b). It is possible that one pK_a reflects deprotonation of a His side chain to free it for ligation to Zn^{2+} , while the higher pK_a reflects the pK_a of Zn^{2+} water/hydroxide, although further work will be required to confirm this. Similar two-step protonation behaviour was observed for other active peptides (Supplementary Fig. 1). The catalytic efficiency of the fibrils does not depend on the initial concentration of the monomeric peptide used for assembly; the reaction is first order in peptide in the concentration range studied (24–200 μM) (Fig. 2d).

Characterization of the active species. Having established that the peptides are catalytically active, we used circular dichroism (CD) and infrared (IR) spectroscopies to show that they adopted a β -sheet conformation in the presence of Zn^{2+} . Also, thioflavin T (ThT) binding and negative stain transmission electron microscopy (TEM) were used to probe fibril formation. The Leu-containing peptide 7 adopts a β -sheet structure in a strictly Zn^{2+} -dependent manner. In the absence of Zn^{2+} the peptide displays a CD spectrum characteristic of random coils, but upon addition of the metal ion a spectrum typical of the cross- β configuration is observed (Fig. 3a). Comparison of spectra at pH 2 and 8 showed that Zn^{2+} -triggering of β -sheet structure occurred

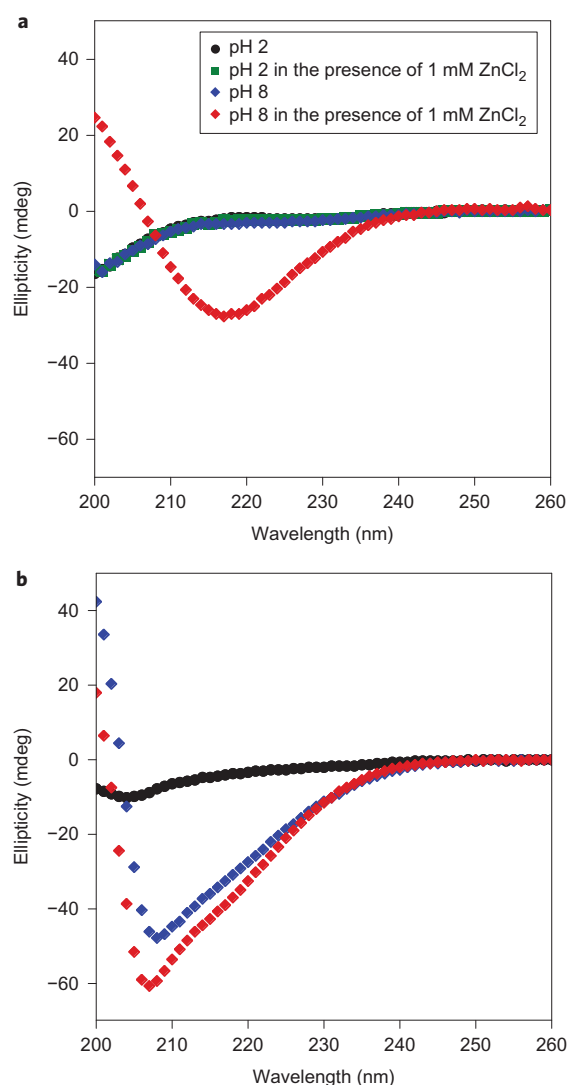


Figure 3 | The active species have a β -sheet secondary structure. a,b, CD spectra of **7** (**a**) and **11** (**b**) under different conditions. In the presence of Zn^{2+} , **7** displays a spectrum typical of the cross- β configuration, and the most active peptide **11** shows a β -sheet-like CD spectrum at pH 8, even in the absence of the metal. Peptide concentration was $24 \mu\text{M}$ in all experiments.

only at pH 8, where the His ligands should be deprotonated and available to complex metal ions. Similar results were seen for the remaining (**1–6**) peptides examined (Supplementary Fig. 2). Interestingly, the nearly inactive peptide **5** (LHLHLHL) showed a β -sheet-like CD spectrum, even in the absence of Zn^{2+} . This finding suggests that this peptide might be pre-organized into a structure that does not support catalysis or, as mentioned above, the additional His competes for catalytically competent Zn^{2+} binding. Interestingly, the most catalytically active Leu-to-Ile substituted peptides (**9**, **11**) showed a β -sheet-like CD spectrum at pH 8, even in the absence of Zn^{2+} (Fig. 3b), suggesting that this substitution promoted the formation of a catalytically competent structure. The position of the amide I' IR band of the catalytically active peptides (**7**, **9**, **11**) near $1,626 \text{ cm}^{-1}$ further supported these conclusions²⁷ (Supplementary Fig. 3). The CD spectra of **9** and **11** at pH 8 in the presence of Zn^{2+} show highly intense minima around 207 nm. Similar spectra were found with fibrils formed by peptides derived from a mouse prion protein and might be indicative of β -structures in readily formed amyloid complexes^{28,29}.

Physical separation methods showed that the active catalyst is assembled into very high-molecular-weight aggregates. Extensive dialysis of **11** using a 10 K molecular weight cut off (MWCO) membrane against zinc-containing buffer showed that the catalytically active species was unable to diffuse through the membrane. Furthermore, when the aggregated peptides were filtered through membranes with 0.1, 0.22 or $0.45 \mu\text{m}$ pores, the filtrate was at least 800-fold less active than the starting suspension (Fig. 4a). Moreover, ThT binding and TEM identified these catalytically active aggregates as fibrils. ThT binds to fibrils in preference to prefibrillar oligomeric states or monomeric peptides, and the binding is conveniently evaluated through a large enhancement in the fluorescence of this dye. Indeed, ThT fluorescence increased significantly in the presence of Zn^{2+} at pH 8, and the most active peptides tended to have the greatest fluorescence enhancement in this assay (Fig. 4b). This finding was confirmed by TEM images of **7**, **9** and **11**, which reveal extended amyloid-like fibrils (Fig. 4c–f, Supplementary Figs 4, 5).

Formation of amyloid-like structures often requires appropriate nucleation and thus a lag phase is frequently observed. We have characterized the activity of designed peptides at various time points after addition of Zn^{2+} and the concomitant pH increase. Peptides **4**, **6** and **7**, which have a Leu core, show a lag phase, reaching their maximum activity at ~ 24 h. Introducing Gln into the sequence and, especially, substituting leucines with isoleucines, drastically diminishes the fibrillation time and the maximum activity is achieved with no need for longer incubation (Supplementary Fig. 6). This observation is consistent with the results of centrifugation studies conducted to determine the fraction of large aggregates. The solutions of representative peptides at pH 8 in the presence of Zn^{2+} were subjected to centrifugation at $100,000g$ for 1 h immediately after preparation and following 24–48 h of incubation at room temperature. Immediately on addition of the pH 8 buffer containing Zn^{2+} , **11** oligomerizes essentially completely (at least 95%), whereas $\sim 55\%$ of peptide **7** is still present in the form of low-molecular-weight species, as judged by their ultraviolet absorbance. After 24 h of incubation, the soluble fraction of **7** drops to $\sim 20\%$ and remains constant thereafter (Supplementary Fig. 7). TEM studies provide additional insight into oligomerization of **7**: freshly produced samples show protofibrils that fully mature after prolonged incubation (Fig. 4c,d). Importantly, non-fibril-forming peptide **8** showed no decrease in soluble fraction absorbance, indicating that no oligomeric species were formed.

Finally, no strong evidence for β -sheet structure based on CD spectra was observed for the catalytically inactive peptide **14**, which has unblocked N- and C-termini. Moreover, under conditions where active peptides showed extensive fibril formation, **14** did not produce fibrils, as judged by TEM. This finding might indicate that the β -strands in the active fibrils are arranged in parallel fashion; in such a scenario, neutralizing the terminal groups by acetylation and amidation would stabilize the structure by eliminating unfavourable electrostatic interactions between charged end groups. However, structural studies will be required to confirm this topology and detailed structure of the fibrils.

Metal coordination sphere. To probe the coordination sphere of the metal ion in the active catalyst we studied the fibrillation of **9** and **11** in the presence of cobalt ions. Co^{2+} triggers fibril formation in a manner analogous to zinc while providing easy spectroscopic insight into the metal coordination geometry. The breadth and intensity of the $d-d$ transition band in the UV-vis spectrum of the fibrils formed by **9** and **11** in the presence of cobalt exclude the possibility of octahedral ligand arrangement and are consistent with tetrahedral cobalt coordination³⁰ (Supplementary Fig. 8). TEM studies show that the morphology of the fibrils formed in the presence of cobalt is not altered compared to the zinc-bound fibrils (Supplementary Fig. 9). These

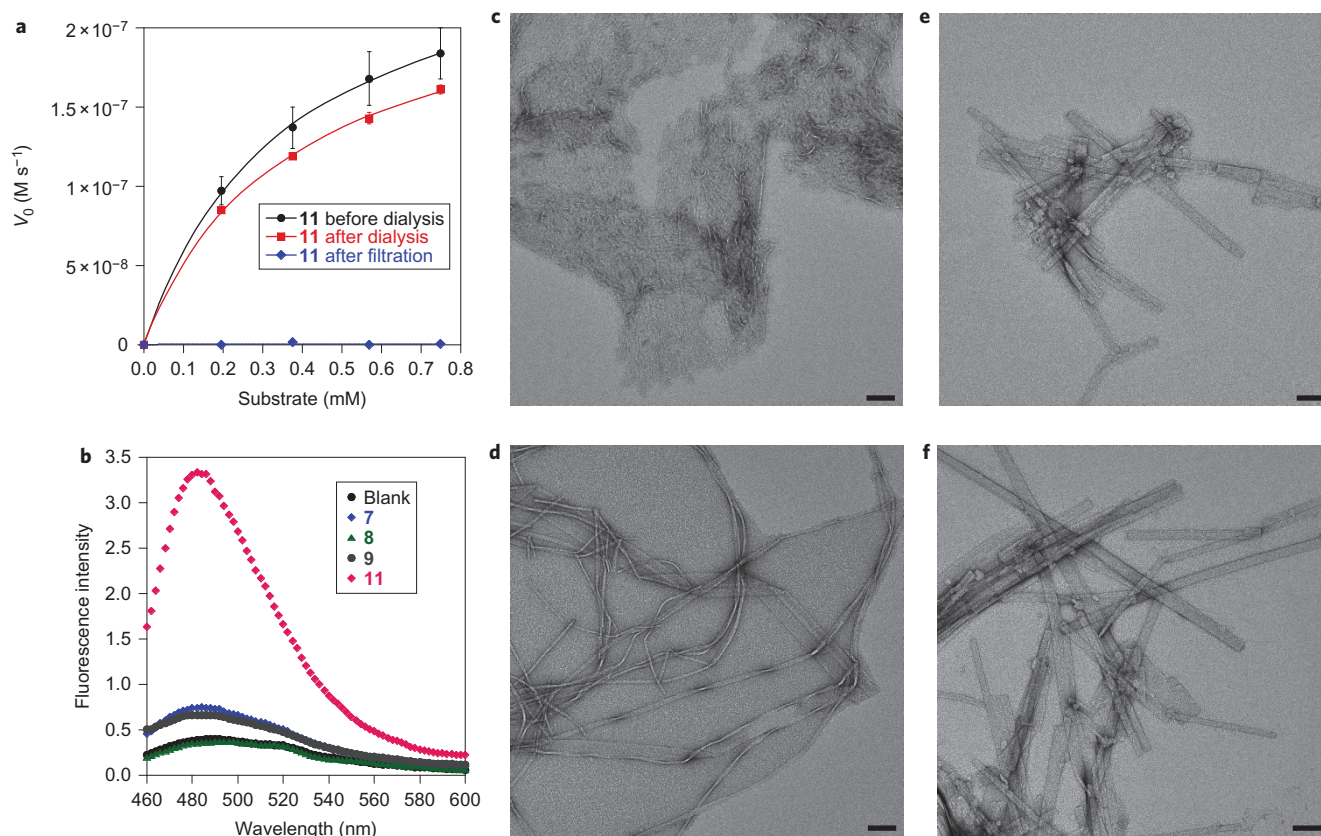


Figure 4 | Large amyloid fibrils are responsible for catalysis. **a**, Extensive dialysis of **11** (24 μ M) does not affect activity (the observed difference in activity is a result of partial irreversible precipitation of the fibrils, as indicated by decreased ultraviolet absorbance). Furthermore, examination of the filtrate after passing an aqueous suspension of **11** through a 450 nm filter shows that the active species are unable to permeate the filter. **b**, Activity correlates with degree of fibrillation of various peptides (**7**, **8**, **9** and **11** as described in the key; peptide concentration was kept at 200 μ M), as shown by thioflavin T (25 μ M) fluorescence increase. **c–f**, Representative TEM images (at $\times 25,000$ magnification; scale bar, 100 nm) of fibrils formed by different peptides at pH 8 in the presence of Zn²⁺: **7** immediately after preparation (**c**); **7** after 72 h incubation at room temperature (**d**); **11** immediately after preparation (**e**); **11** after 72 h incubation at room temperature (**f**). Both **7** and **11** form amyloid-like fibrils. Freshly produced samples of **7** show protofibrils that mature fully after prolonged incubation, whereas **11** forms fibrils very quickly, essentially upon mixing. Peptide concentration was 24 μ M in all TEM experiments. Error bars in all graphs represent standard deviations of at least three independent measurements.

findings are in good agreement with the results of computational modelling of the assembly using Rosetta software (see Supplementary Information, page 5 for details). The energy-minimized model of the fibrils formed by **11** in the presence of Zn²⁺ ions is presented in Fig. 1c–e. It shows a well-packed hydrophobic core formed by isoleucine residues, where the histidine residues support a tetrahedral zinc coordination sphere.

Synergistic interactions between peptides. The above studies show that relatively simple design principles are needed to discover amyloid-like peptides that display very substantial catalytic activity for ester hydrolysis. More than half of the 14 peptides studied here were active, and the activity could be easily enhanced by following simple physical principles such as enhancing the amyloidogenic potential and β -sheet-forming properties of the peptides. This is but a small fraction of the $20^7 = 1.28 \times 10^9$ possible peptide sequences that could be composed of the 20 naturally occurring amino acids, or $3^7 = 2,187$ if one were to restrict the alphabet to just three amino acids, such as His, Ile and Gln in the most active peptides studied here. It remains to be seen how many other activities could be discovered from relatively simple libraries in the absence and presence of other biological and non-biological metal ions. Furthermore, diversity could be greatly enhanced (by a factor of $(n + n^2)/2$, where n is the number of peptides in the library) if the peptides were considered in

binary mixtures of only two peptides. To test the possibility that a mixture of two peptides might have synergistic interactions we mixed the two peptides that showed moderate activity in the initial screening (**4** and **7**). To assure mixing of the individual peptides within the fibril, they were mixed in different proportions in aqueous HCl (10 mM) and then diluted into Zn²⁺-containing (1 mM) Tris buffer at pH 8 to ensure effective mixing of peptides before the formation of fibrils. The activity profile shows synergy between **4** and **7** (Fig. 5a) with maximum esterase activity observed when the peptides were mixed relative to the individuals alone. The experimental data are well described by a simple model, where a dimer is the active species and the peptides are randomly distributed in the fibril. By contrast, if fibrils of the individual peptides were allowed to form first, and then they were mixed, no synergy was observed. Instead a linear decrease in activity was seen upon increasing the ratio of **4** relative to **7** (Fig. 5b). Detailed structural studies will be necessary to elucidate the composition and geometry of the active unit as the data could also be fit to more complex oligomerization (e.g. peptide trimer constituting the active species) models.

Conclusions

Here, we have shown that small seven-residue amyloid-forming peptides designed from first principles form efficient catalysts of ester hydrolysis. This observation supports a potential link

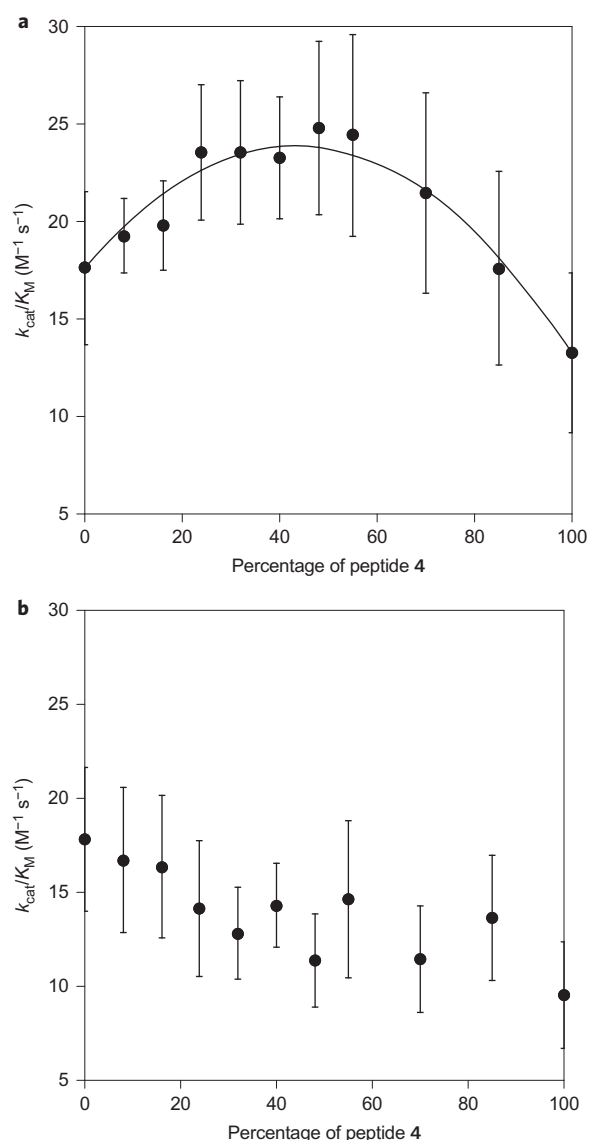


Figure 5 | Synergistic interactions between peptides. **a**, The catalytic efficiency (k_{cat}/K_M) profile shows synergy between **4** and **7** with maximum esterase activity observed when the peptides were mixed relative to the individuals alone. The experimental data are well described by a simple model, where a dimer is the active species and the peptides are randomly distributed in the fibril (fit). **b**, No synergy is observed when preformed fibrils of **4** and **7** are mixed together; instead, a linear decrease in activity is seen upon increasing the ratio of **4** relative to **7**. Conditions: pH 8; total peptide concentration kept at 24 μM ; buffer containing 1 mM Zn^{2+} ; fibrils equilibrated for 48 h before measurements. Error bars in all graphs represent standard deviations of at least three independent measurements.

between amyloid assemblies and the emergence of protein catalysts during the evolution of enzymes. The ability to screen multiple stable arrangements of functional groups in a single fibril provides essentially limitless opportunities for high-throughput screening for functional activity by simply mixing peptides with different sequences. We expect this approach to form a tool for designing new materials capable of performing catalysis on both natural and non-natural substrates. This work might also have implications for the mechanisms of amyloid toxicity, with zinc and copper ions altering the kinetics and nature of the cellular prion protein (PrP^C)^{31,32}. Moreover, it has been suggested that amyloid-forming A β -peptides might form complexes with Cu²⁺ that mediate the

formation of reactive oxygen species (ROS) in an incompletely understood manner^{33–35}. The ease with which we were able to discover robust and reproducible metalloprotein catalysts would suggest that toxic catalytic activities should be considered as one possible source of amyloid toxicity.

Methods

Peptide synthesis. The peptides were synthesized by manual fluorenylmethyloxycarbonyl (Fmoc) solid-phase synthesis at elevated temperature using our previously reported protocol optimized for hydrophobic peptides³⁶. Cleavage of the peptides from resin and side-chain deprotection was simultaneously achieved by treatment with a mixture of trifluoroacetic acid (TFA)/H₂O/triisopropyl silane (TIS) (95:2.5:2.5, vol/vol) for 2 h at room temperature. The crude peptides were precipitated and washed with cold methyl-*tert*-butyl ether, and purified on a preparative reverse-phase high performance liquid chromatography (HPLC) system (Varian ProStar 210) with a C4 preparative column (Vydac), using a linear gradient of solvent A (0.1% TFA in Millipore H₂O) and solvent B (90% CH₃CN, 10% H₂O, 0.1% TFA). The identities and purities of the purified peptides were confirmed by matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry using a Bruker Autoflex III Smartbeam MALDI-TOF mass spectrometer. Purity of the obtained peptides was additionally evaluated on a Shimadzu Prominence UFLC instrument with an analytical Zorbax Eclipse XDB-C18 column (4.6 mm $\times$ 150 mm).

Preparation of peptide stocks and solutions. Purified lyophilized peptides were dissolved in 10 mM hydrochloric acid to make a 1.1 mM stock solution. Concentration was determined using an Agilent 8453 UV-Vis spectrophotometer using absorbance at 214 nm. Extinction coefficients for the peptides at this wavelength were calculated using literature values³⁷. The pH 8 working stock solution of peptides was prepared by mixing 180 μl of the pH 2 stock with 20 μl isopropanol followed by addition of 1.8 ml buffer (25 mM Tris pH 8 containing 1 mM ZnCl₂). Adherence to this protocol assured reproducibility between runs and provided a standard method to compare different peptides. The pH 2 stock solution was stable for at least a week, while the working stock solution of peptides at pH 8 was prepared immediately before experiments.

Kinetic assays. Kinetic measurements were done on a Thermo Labsystems Multiskan Spectrum plate reader monitoring absorbance of the product (*p*-nitrophenol) at 348 nm and 405 nm at 22 $^{\circ}C$ in 96-well plates. *p*-Nitrophenylacetate was used from a 0.1 M stock in acetonitrile (the final acetonitrile content was 2% in all reaction mixtures). A volume of 150 μl of freshly prepared substrate solution in 25 mM Tris (pH 8), 1 mM ZnCl₂ (to a final substrate concentration of 0.195–0.75 mM) was added to 50 μl buffered (25 mM buffer) peptide stock solution at pH 8 (to a final peptide concentration of 24 μM). Extinction coefficients of 1,700, 4,300, 9,100, 12,700 and 16,600 $M^{-1} cm^{-1}$ were used to determine the product concentration at pH values of 6, 6.5, 7, 7.5 and 8, respectively, at 405 nm. An extinction coefficient of 18,700 $M^{-1} cm^{-1}$ was used at pH 8.5 and above at 405 nm. The extinction coefficients reported above were experimentally obtained by measuring UV-vis spectra of *p*-nitrophenol in buffers at different pH values. The *p*-nitrophenol solutions were obtained by 40-fold dilution of 2.5 mM stock solution of *p*-nitrophenol in acetonitrile into the appropriate buffers. The reported results were obtained by averaging at least three independent measurements. Kinetic parameters (k_{cat} and K_M) were obtained by fitting the data to the Michaelis–Menten equation ($v_0 = k_{cat}[E]_0[S]_0/(K_M + [S]_0)$). k_{cat}/K_M values were obtained by fitting the linear portion of the Michaelis–Menten plot to $v_0 = (k_{cat}/K_M)[E]_0[S]_0$. pH profile studies were performed on an Agilent 8453 spectrophotometer monitoring absorbance of the product at 405 nm at 22 $^{\circ}C$ using a 1 cm quartz cuvette. A volume of 150 μl buffer solution (the following buffers were used at a concentration of 25 mM: MES (pH 6–6.5), HEPES (pH 7–7.5), Tris (pH 8–8.5), TAPS (pH 9–9.5) and CAPS (pH 10–10.5)) was added to 50 μl peptide solution (in 25 mM Tris pH 8 containing 1 mM ZnCl₂) in a cuvette, followed by the addition of 2 μl of 20 mM pNPA in acetonitrile. Because of the precipitation of zinc hydroxide at high pH values, the added buffers at pH values of 8 and below contained 1 mM ZnCl₂, the buffers at pH 8.5–9 contained 0.5 mM ZnCl₂ (final Zn²⁺ of 0.625 mM) and at pH values above 9.5 the added buffer contained no zinc salts (final Zn²⁺ of 0.25 mM). pH changes arising from mixing different buffer volumes at different pH values were taken into account. Product formation was monitored for 5 min every 30 s. The pH profile data were fit to the following equation:

$$\frac{k_{cat}}{K_M} = \frac{\left(\frac{k_{cat}}{K_M}\right)_{\max 1} 10^{-pK_{a1}}}{10^{-pH} + 10^{-pK_{a1}}} + \frac{\left(\frac{k_{cat}}{K_M}\right)_{\max 2} 10^{-pK_{a2}}}{10^{-pH} + 10^{-pK_{a2}}}$$

where k_{cat}/K_M is the observed enzymatic efficiency at a particular pH value, pK_{a1} and pK_{a2} are the respective pK_a values for the two protonation/deprotonation steps and $(k_{cat}/K_M)_{\max 1}$ and $(k_{cat}/K_M)_{\max 2}$ are the corresponding maximum activities.

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Author contributions

C.M.R., Y.S.M., O.V.M., J.S., T.A.S. and X.H. performed the experiments and analysed the data. W.F.D. and I.V.K. conceived and designed the experiments and wrote the manuscript. All authors discussed the results and commented on the manuscript.

Additional information

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Competing financial interests

The authors declare no competing financial interests.